

TACHS Exam Simulator

Full-Length Practice Exam 04

Complete printable test with 200 multiple-choice questions, answer key, and detailed explanations.

Section	Questions	Time
Reading	50	40 minutes
Written Expression	50	40 minutes
Mathematics	50	45 minutes
Ability	50	35 minutes

General Directions

Work through each section within the recommended time. Choose the best answer for every question. Mark one answer (A-D) for each item. Use the answer key and explanations only after completing the test.

Recommended total testing time: 160 minutes. Calculators should be used only when permitted by the rules followed for your TACHS preparation.

Section 1: Reading

50 questions | Recommended time: 40 minutes

Literary Passage: The Missing Measure

For weeks, Jonah had practiced a short piano duet with his cousin Elise. At the community recital, they would play on the hall's old upright piano, whose keys felt heavier than those on Jonah's keyboard at home. He worried about the quick final section, so he practiced it until his hands could nearly move without thought.

Just before the recital, Elise opened the music and discovered that the last page was missing. Jonah searched their folders, the bench, and even beneath the piano. The audience settled into its seats while the program director quietly asked whether they needed to skip their turn.

Elise shook her head. She remembered the upper part, and Jonah knew most of the lower one. Still, he could not recall two measures near the end. Then he noticed a pattern: the uncertain measures echoed an earlier phrase, but began one step lower. He traced the pattern on the silent keys and nodded.

They began. The unfamiliar weight of the keys demanded attention, yet Jonah listened closely to Elise instead of racing ahead. When they reached the missing measures, the repeated pattern guided him. Their final chord was not perfectly simultaneous, but it was strong. Backstage, Jonah found the missing page tucked inside another piece. He laughed. The page had been useful during practice, but careful listening and understanding the music had carried them through the performance.

1. Which statement best expresses the central idea of the passage?

- A) Understanding patterns and listening carefully help Jonah handle an unexpected problem.
- B) Musicians should always memorize every page before performing.
- C) An old piano prevents Jonah and Elise from playing well together.
- D) The program director wants the cousins to skip their recital.

2. Where was the missing page eventually found?

- A) Beneath the piano
- B) Inside another piece of music
- C) In the program director's folder
- D) Under a seat in the audience

3. What can reasonably be inferred about Jonah during the performance?

- A) He ignores Elise in order to concentrate on the keys.
- B) He decides that the final section is too difficult to attempt.
- C) He adjusts his approach instead of depending only on memorized movements.
- D) He discovers that the hall's piano is easier to play than his keyboard.

4. How does the discovery of the repeated musical phrase affect the plot?

- A) It explains why the music page was placed in the wrong folder.
- B) It creates a disagreement between Jonah and Elise.
- C) It causes the program director to delay the recital.
- D) It provides Jonah with a way to reconstruct the measures he cannot remember.

Informational Passage: When a Lake Turns Over

In many deep lakes, the water does not remain evenly mixed throughout the year. During summer, sunlight warms the upper layer. Because warm water is less dense than cold water, it stays near the surface. Deeper water remains cooler, and a narrow transition zone separates the two layers. Wind may stir the surface, but it usually cannot mix the entire lake.

As autumn air cools the surface, the upper water becomes denser. Eventually, its temperature approaches that of the deeper water. The density difference weakens, allowing wind to circulate water from top to bottom. This seasonal mixing is called turnover.

Turnover redistributes more than heat. During summer, tiny organisms near the surface may use dissolved nutrients, while decaying material carries nutrients toward the bottom. At the same time, deep water may receive little new oxygen from the air. When the lake mixes, oxygen moves downward and nutrients move upward. This process can briefly make the water look cloudy because sediment and organic material are stirred. The cloudiness does not necessarily indicate pollution. In fact, turnover often supports the lake's living community by renewing access to oxygen and nutrients. A similar mixing period may occur in spring, after ice melts and before surface water becomes much warmer than deep water.

5. Why does warm surface water remain above colder water during summer?

- A) Warm water is less dense than cold water.
- B) Wind pushes all cold water to the lake bottom.
- C) Dissolved nutrients make surface water lighter.
- D) Sediment prevents the two layers from touching.

6. As used in the third paragraph, “redistributes” most nearly means

- A) measures for the first time
- B) spreads into new locations
- C) removes completely
- D) changes into another substance

7. Why does the author mention that turnover may make lake water appear cloudy?

- A) To argue that cloudy lakes should be closed to visitors
- B) To show that sediment is the main cause of lake pollution
- C) To clarify that a visible change can be part of a beneficial natural process
- D) To prove that autumn turnover is stronger than spring turnover

8. Which condition would most likely delay autumn turnover?

- A) Nutrients collecting near the lake bottom
- B) A brief increase in wind at the surface
- C) Organisms using oxygen in deep water
- D) Surface water remaining much warmer than deep water

Humanities Passage: The Work Behind a Useful Map

A map may appear to be a simple picture of a place, but every map is also a collection of decisions. A cartographer must decide what to include, what to leave out, and how to represent features that cannot all fit at their actual size. Those decisions depend on purpose. A subway map emphasizes station order and transfer points, even if the drawn routes bend differently from the tracks beneath the city. A hiking map gives more attention to elevation, water sources, and trail junctions. A map made for emergency planning may highlight evacuation routes and shelters while omitting details that would distract from quick decisions.

Scale creates another challenge. On a neighborhood map, individual streets and buildings may be useful. On a map of an entire country, those same features would crowd the page. Cartographers therefore generalize: they simplify coastlines, combine minor roads, or select only the most relevant landmarks.

These choices do not make a carefully designed map dishonest. Instead, they remind readers to ask what job the map was designed to perform. Trouble begins when a reader expects one map to answer every possible question. The best map is not necessarily the one with the most information. It is the one whose selected information helps its intended users see the relationships they need.

9. Which statement best summarizes the passage?

- A) Effective maps select and simplify information according to a specific purpose.
- B) Subway maps are less accurate than maps used by hikers.
- C) Cartographers should include every landmark that can fit on a page.
- D) Map readers mainly need to understand how scale is calculated.

10. In the third paragraph, “generalize” most nearly means

- A) guess without evidence
- B) present in a simplified form
- C) enlarge every small feature
- D) arrange in alphabetical order

11. According to the passage, what does a subway map emphasize?

- A) The exact curves followed by underground tracks
- B) The elevation of each neighborhood
- C) Station order and transfer points
- D) Every street located above the subway

12. Which conclusion is best supported by the passage?

- A) A detailed map is useful for every type of traveler.
- B) Simplifying a map always makes it less trustworthy.
- C) Emergency maps should show the locations of all buildings.
- D) A map should be judged partly by how well it serves its intended users.

13. The author’s primary purpose is to

- A) explain why mapmaking requires purposeful selection
- B) persuade readers to avoid using digital maps
- C) compare the careers of several cartographers
- D) describe how to measure distance with a map scale

Literary or Narrative Passage: The Reversed Poster

Leah volunteered to print a large welcome poster for the school language fair. The art room had a small hand press, and she arranged wooden letters in a frame exactly as the words should appear. When she lifted the first sheet, the message was backward.

Mr. Okafor did not reset the letters for her. Instead, he placed a mirror beside the frame. In the reflection, Leah could read the arrangement correctly. She realized that the raised letters touched the paper face down, so the frame had to hold a reversed version of the final message.

Leah began again, checking each line in the mirror before tightening the frame. The second proof was readable, but the spaces between words were uneven. She inserted thin blank blocks, printed another proof, and compared the margins on both sides. Each test used more paper, yet every proof revealed a different problem while it was still easy to fix.

On the morning of the fair, the finished poster hung above the entrance. Visitors saw a clean message in three languages. Leah noticed something else: the poster looked simple only because the complicated decisions had been settled before the final sheet was printed.

14. What tool helps Leah check whether the letters are arranged correctly?

- A) A ruler beside the press
- B) A mirror beside the frame
- C) A lamp beneath the paper
- D) A photograph of the entrance

15. Why does Leah print several proofs before making the final poster?

- A) She wants to display every version at the fair.
- B) She must use all the paper in the art room.
- C) She is unsure which languages belong on the poster.
- D) Each proof reveals a problem that can still be corrected.

16. Which statement best expresses the main idea of the passage?

- A) A polished result often depends on careful testing and correction.
- B) Hand presses are faster than modern printers for large posters.
- C) School fairs should display information in several languages.
- D) Mirrors are the most important tools in an art room.

17. As used in the third paragraph, “proof” most nearly means

- A) evidence that an argument is true.
- B) a protective coating placed on paper.
- C) a trial copy made for checking.
- D) a rule that cannot be changed.

Scientific or Technical Passage: Why Owl Wings Sound Quiet

Many birds create a noticeable rush of air as their wings move. Owls, however, can approach prey with unusually little flight noise. Their silence does not come from one feature alone. Several parts of an owl feather work together to change how air passes over the wing.

Along the front edge of a primary wing feather, small comb-like projections divide a strong stream of air into many smaller currents. A soft, velvety surface on the feather absorbs some of the tiny vibrations created by those currents. The trailing edge is fringed rather than sharply cut, which allows air to leave the wing more gradually.

These features reduce noise, but they also create trade-offs. The soft feather surface is not ideal for high-speed flight in heavy rain, and most owls do not travel as rapidly as some daytime hunters. Quiet movement is especially useful during short flights when an owl must listen for faint sounds below.

Engineers study owl feathers when designing quieter fans, turbines, and aircraft parts. A useful design does not simply copy the shape of a feather. It identifies how separated airflow, soft surfaces, and gradual edges reduce sound, then adapts those principles to a different material and purpose.

18. What do the comb-like projections on an owl feather do?

- A) Divide a strong air stream into smaller currents
- B) Hold rainwater away from the wing
- C) Increase the bird's speed during long flights
- D) Make the feather surface harder

19. Why does the author mention fans, turbines, and aircraft parts?

- A) To show that owl feathers are made from industrial materials
- B) To argue that machines should move as slowly as owls
- C) To explain how biological principles can inspire quieter technology
- D) To prove that all flight noise can be eliminated

20. How is the passage mainly organized?

- A) It compares several owl species and ranks their hunting success.
- B) It gives instructions for building a model wing.
- C) It presents a mystery and then rejects every possible explanation.
- D) It explains several feather features, their effects, and possible applications.

21. As used in the third paragraph, "trade-offs" most nearly means

- A) exchanges of feathers between birds.
- B) advantages gained with accompanying disadvantages.
- C) changes that occur only during rain.
- D) measurements used to compare flight speed.

Informational or Historical-Style Passage: When Elevators Learned to Show Their Location

Early passenger elevators moved people between floors, but riders often had little information about where the car was. An operator controlled the motion and announced each stop. In a busy building, a person waiting in the hallway might not know whether the elevator was one floor away or near the roof.

As buildings grew taller, manufacturers added position indicators. Some early systems used a moving pointer beside a vertical list of floor numbers. Others placed small lamps above the doors. The elevator car activated a switch as it passed each floor, causing the matching lamp to glow.

The indicator did more than satisfy curiosity. People waiting could decide whether to remain at one elevator or walk to another. Maintenance workers could also notice when the displayed position did not match the car's actual location. Later systems added direction arrows, arrival signals, and spoken announcements.

Modern digital displays look different from the mechanical pointers of earlier buildings, but they solve the same communication problem. The machine is moving inside a closed shaft, while the people using it need clear information outside that shaft. A small display connects those two situations by making hidden motion understandable.

22. How did some early elevator indicators show the car's position?

- A) By opening a window into the elevator shaft
- B) By printing a new ticket at each floor
- C) By moving a pointer or lighting a floor lamp
- D) By slowing the car near every hallway

23. Why did position indicators become more useful as buildings grew taller?

- A) The elevator could be farther away and harder to track without information.
- B) Tall buildings required every rider to operate the elevator.
- C) Mechanical pointers worked only above a certain height.
- D) People in tall buildings were not allowed to use stairs.

24. What is the author's main purpose in the final paragraph?

- A) To criticize digital displays for replacing mechanical devices
- B) To show that old and new indicators address the same need for information
- C) To explain how to repair an elevator position switch
- D) To argue that elevator shafts should be visible

25. Which statement best summarizes the passage?

- A) Elevator operators disappeared as soon as electric lamps were invented.
- B) Tall buildings were designed mainly to test new communication devices.
- C) Mechanical pointers are more accurate than modern digital displays.
- D) Position indicators made hidden elevator movement clearer and more useful to people.

Literary Passage: The Name Beside the Photograph

Eli volunteered to help prepare a small history exhibit for the neighborhood library. Most of the photographs already had labels, but one image showed a young woman standing beside a delivery bicycle with no name beneath it. The only clue was a penciled note on the back: "First route, 1948."

The exhibit coordinator suggested labeling the woman as an unidentified worker. Eli hesitated. The bicycle carried the faded emblem of a bakery that had once operated two blocks away, so he searched old advertisements and city directories. He found the bakery's address but no list of employees. Next, he compared the photograph with a group picture from a street festival. One face looked similar, although the image was too blurred to be certain. On opening day, Eli placed a temporary card beside the photograph explaining what he had learned and asking visitors to share information. An older visitor studied the image for a long time. She recognized the woman as her aunt, Celia Moreno, who had delivered bread while the bakery's regular drivers were serving overseas. The visitor returned the next morning with a letter that mentioned Celia's route.

Eli replaced the temporary card with a fuller label, including Celia's name and the source of the identification. The coordinator thanked him for refusing to make a quick guess. Eli realized that leaving a question open had not weakened the exhibit. It had invited the community to help complete the record.

26. Which choice best states the central lesson developed through Eli's work on the exhibit?

- A) Historical exhibits should include only photographs with complete labels.
- B) Old business records are more reliable than family memories.
- C) Careful research and community knowledge can restore an overlooked part of history.
- D) Libraries should avoid displaying photographs from private collections.

27. What information was written on the back of the photograph?

- A) "First route, 1948."
- B) The bakery owner's name
- C) A list of delivery addresses
- D) The date of the street festival

28. Why did Eli use a temporary card at first?

- A) He wanted the photograph removed after opening day.
- B) He had forgotten to print the permanent exhibit labels.
- C) He believed visitors would prefer a shorter description.
- D) He wanted to share verified facts without pretending the woman's identity was certain.

29. As used in the final paragraph, "invited" most nearly means

- A) formally requested attendance from.
- B) encouraged participation from.
- C) provided shelter for.
- D) gave instructions to.

Informational Passage: Why Some Rooms Echo

An echo occurs when sound waves strike a surface and return to a listener after a noticeable delay. In a large, empty room, hard walls, floors, and ceilings reflect much of the sound instead of absorbing it. A clap may therefore be heard more than once, or speech may seem to linger after a person stops talking.

Not every reflected sound becomes a distinct echo. In smaller rooms, reflections often return so quickly that they blend with the original sound. This blending is called reverberation. A little reverberation can make music sound rich, but too much can make words difficult to understand because one syllable overlaps the next.

Designers control reflections in several ways. Curtains, upholstered seats, carpets, and acoustic panels absorb part of a sound wave's energy. Irregular surfaces can diffuse sound, scattering it in many directions rather than sending a strong reflection toward one location. The placement of these materials matters as much as the amount used.

The goal is usually balance, not complete silence. A lecture hall needs clear speech, while a concert hall may benefit from longer reverberation. By measuring how quickly sound fades and where strong reflections occur, designers can shape a room for its intended use.

30. According to the passage, what is reverberation?

- A) A sound that is completely absorbed by soft material
- B) Reflected sounds that return quickly and blend with the original sound
- C) A vibration produced only by musical instruments
- D) A distinct echo heard after a long period of silence

31. Which description best matches the passage's overall structure?

- A) It compares the construction costs of several kinds of rooms.
- B) It presents a historical problem and describes who first solved it.
- C) It lists materials in the order that they should be installed.
- D) It explains sound reflection, distinguishes related effects, and describes design responses.

32. Why does the author contrast a lecture hall with a concert hall?

- A) To show that the desired amount of reverberation depends on a room's purpose
- B) To prove that music is always louder than speech
- C) To argue that lecture halls should contain no hard surfaces
- D) To explain why concert halls are usually larger

33. As used in the third paragraph, "diffuse" most nearly means

- A) measure precisely.
- B) block completely.
- C) scatter in several directions.
- D) increase in volume.

Humanities Passage: The Many Jobs of a Public Square

Public squares have served cities for centuries, but their value does not come from one fixed design. Some began as marketplaces where farmers and craftspeople sold goods. Others formed near government buildings, places of worship, or transportation routes. Their locations made them natural meeting points.

A successful square usually provides both open space and useful edges. The center may hold a market, performance, or public ceremony. Along the sides, trees, steps, benches, and storefronts give people places to pause. These boundaries help a large space feel connected to everyday street life rather than empty or exposed. The uses of a square can change throughout a single day. Commuters may cross it in the morning, children may play there in the afternoon, and an audience may gather for music in the evening. Designers cannot predict every activity, so movable chairs, level surfaces, shade, and access from several directions can make the space adaptable.

When a square is judged only by its appearance in a photograph, its most important quality may be missed. A public square succeeds when people can enter it easily, remain comfortably, and use it in more than one way. Its design is therefore less like a finished picture than a framework for shared life.

34. Which choice most accurately summarizes the passage about public squares?

- A) Why modern cities no longer need traditional marketplaces
- B) How government buildings determine the shape of nearby streets
- C) Why public ceremonies should be held in open spaces
- D) How flexible design allows public squares to support many forms of shared activity

35. According to the passage, what can help the edge of a square feel connected to street life?

- A) A single large monument in the center
- B) Trees, steps, benches, and storefronts
- C) Fences that restrict entry after dark
- D) Permanent seating arranged in straight rows

36. Which feature would the author most likely favor in a newly designed square?

- A) A decorative surface that cannot support events
- B) One narrow entrance that controls all movement
- C) Movable seating that can be rearranged for different uses
- D) A central area reserved permanently for one activity

37. Why does the author compare a public square to “a framework for shared life”?

- A) To emphasize that the space should support changing activities rather than function as a fixed display
- B) To suggest that every square should contain the same structural materials
- C) To criticize architects for paying attention to appearance
- D) To show that squares are less durable than surrounding buildings

38. As used in the third paragraph, “adaptable” most nearly means

- A) historically important.
- B) visually impressive.
- C) protected from weather.
- D) able to serve changing needs.

Literary or Narrative Passage: The Stubborn Pulley

Nico was helping the stage crew raise a painted backdrop for the school play. During rehearsal, the backdrop stopped halfway up and tilted toward one side. Pulling harder only made the rope groan.

Ms. Patel asked everyone to step back. She traced the rope from the handle to the ceiling and noticed that one pulley wheel barely turned. Dust had collected around its axle, and the rope had slipped toward the rim. Nico suggested spraying oil from the floor, but Ms. Patel explained that too much oil could drip onto the fabric and attract more dust.

After the ladder was secured, she cleaned the axle with a narrow brush and applied one small drop of lubricant. Nico guided the rope into the center groove while another student kept the backdrop supported. They tested the system with a sandbag before attaching the painted cloth again.

This time the backdrop rose evenly. Nico had expected the solution to require a stronger pull or a new rope. Instead, the repair depended on reducing resistance at one small point. During the performance, the audience applauded the scenery change. The crew listened for the pulley and heard almost nothing.

39. What problem does Ms. Patel find in the pulley system?

- A) The painted cloth is too heavy for the rope.
- B) The handle is attached to the wrong wall.
- C) One pulley wheel barely turns and the rope has shifted toward its rim.
- D) A sandbag is caught above the backdrop.

40. Why does the crew test the pulley with a sandbag before reattaching the backdrop?

- A) To confirm that the repair works without risking the painted cloth
- B) To make the rope stretch before the performance
- C) To clean dust from the ceiling
- D) To show the actors how scenery is moved

41. Which statement best expresses a central idea of the passage?

- A) Stage scenery should always be moved by adults.
- B) A new rope solves most pulley problems.
- C) Audiences notice every backstage repair.
- D) A small source of resistance can matter more than additional force.

42. As used in the third paragraph, “secured” most nearly means

- A) decorated.
- B) made stable and safe.
- C) hidden from view.
- D) measured for height.

Informational or Scientific Passage: How Antlion Pits Trap Moving Sand

An antlion larva lives beneath loose, dry sand and builds a cone-shaped pit. It begins by moving backward in a circle while flicking grains away with its head. As the circle tightens, the center becomes deeper and the sides become steep.

The slope of the finished pit is close to the steepest angle at which the sand can remain stable. When a small insect steps onto the edge, grains begin to slide. The insect tries to climb, but each movement may loosen more sand. The antlion waits below the center and sometimes throws additional grains upward, creating more small slides.

The pit does not pull an insect downward like a machine. Instead, it uses the behavior of loose grains. A shallow slope would be easier to cross, while a much steeper slope would collapse before the trap was ready. The antlion repeatedly repairs the sides so they remain near the useful angle.

Scientists study these pits to understand movement on unstable surfaces. The same principles can help explain why feet, wheels, and digging tools behave differently in sand than on solid ground. The antlion's trap is effective because its shape works with the material rather than against it.

43. How does an antlion begin building its pit?

- A) By carrying wet sand toward the center
- B) By moving backward in a circle and flicking grains away
- C) By digging a straight tunnel beneath a stone
- D) By pressing the sand into a hard wall

44. Why does the author explain that the pit does not pull an insect downward like a machine?

- A) To show that antlions are too weak to move sand
- B) To compare the pit with a mechanical elevator
- C) To argue that machines cannot work on loose ground
- D) To emphasize that the trap depends on the behavior of unstable grains

45. How does the third paragraph contribute to the passage?

- A) It explains why the pit must have a carefully balanced slope.
- B) It lists several animals that escape antlion pits.
- C) It gives instructions for keeping an antlion as a pet.
- D) It describes how rain fills the pit with water.

46. As used in the final paragraph, "principles" most nearly means

- A) official rules for protecting insects.
- B) tools used to measure the pit.
- C) basic ideas that explain how something works.
- D) steps performed in a fixed order.

Humanities or Reflective Passage: Why Translators Keep More Than One Draft

Translating a paragraph involves more than replacing each word with a word from another language. A sentence may contain an expression that sounds natural to one audience but confusing to another. Its rhythm may also affect whether the speaker sounds formal, playful, impatient, or calm.

For this reason, translators often prepare several drafts. One draft may stay close to the original sentence structure so that no idea is lost. A second may rearrange the wording to sound natural in the new language. A third may focus on tone, especially when the text includes humor, dialogue, or instructions.

The drafts are then compared. A smooth sentence is not successful if it changes the meaning, and a perfectly literal sentence may fail if readers cannot understand its purpose. Translators may read a passage aloud, ask another speaker to review it, or return to surrounding paragraphs for context.

The final version hides most of this comparison. Readers see one set of words, not the abandoned choices beside it. Yet those alternatives are valuable. They allow the translator to test meaning, clarity, and voice separately before selecting language that balances all three.

47. What may a translator's second draft do according to the passage?

- A) Remove every difficult sentence
- B) Copy the original word order exactly
- C) Add facts not included in the original
- D) Rearrange wording so it sounds natural in the new language

48. Why might a translator read a passage aloud?

- A) To replace the written version with a recording
- B) To judge how naturally the wording and tone sound
- C) To avoid comparing the passage with surrounding paragraphs
- D) To count the number of words in each sentence

49. What does the author emphasize by saying that the final version "hides" the comparison?

- A) Translators should never share early drafts.
- B) Readers are unable to understand translated writing.
- C) A finished translation may conceal extensive testing of alternatives.
- D) Only the shortest draft should be published.

50. Which statement best summarizes the passage?

- A) Multiple drafts help translators balance meaning, clarity, and tone.
- B) Literal translations are always more accurate than natural ones.
- C) Every translated paragraph should preserve the original word order.
- D) Translators mainly revise a text to make it shorter.

Section 2: Written Expression

50 questions | Recommended time: 40 minutes

Revision Passage: A Hallway Gallery

[1] Students at Monroe Middle School wanted more people to notice the artwork created in their classes. [2] They proposed turning an empty hallway into a rotating gallery. [3] First, the group measured the walls and asked the principal which hanging materials were permitted. [4] The cafeteria serves soup on Wednesdays. [5] Next, students designed reusable labels that listed each artist's name, medium, and brief statement. [6] Because the hallway leads to the auditorium, families can view the display before concerts and plays. [7] A new set of work will be installed every six weeks. [8] The gallery gives student artists a public audience while keeping the display fresh throughout the year.

51. Which sentence should be removed because it does not support the passage's central idea?

- A) Sentence 3
- B) Sentence 4
- C) Sentence 6
- D) Sentence 7

52. Which revision most effectively combines sentences 1 and 2?

- A) Students at Monroe Middle School wanted more people to notice their artwork, turning an empty hallway.
- B) Wanting artwork, an empty hallway was proposed by students as a rotating gallery.
- C) To help more people notice student artwork, students at Monroe Middle School proposed turning an empty hallway into a rotating gallery.
- D) Students wanted more people to notice artwork, they proposed an empty hallway gallery.

53. Which word best replaces "brief" in sentence 5 without changing the intended meaning?

- A) concise
- B) tiny
- C) hurried
- D) narrow

54. Which sentence would best follow sentence 7?

- A) Some students prefer painting, while others like sculpture.
- B) The school auditorium contains several rows of folding seats.
- C) Labels can be printed in more than one typeface.
- D) This schedule will allow many classes to share the limited wall space.

Revision Passage: Testing a Shade Sail

[1] The science club wondered whether a fabric shade sail could lower the temperature above the school's sunny courtyard tables. [2] Members placed identical thermometers above two tables at the same height. [3] One table was covered by the sail, while the other remained in direct sunlight. [4] The club recorded both temperatures every ten minutes for one hour. [5] Clouds covered the sun for several minutes during the test. [6] As a result, members repeated the experiment on two additional clear days. [7] The shaded table was cooler during most readings on all three clear-day trials. [8] The club presented its results to the principal and recommended a longer test in different seasons.

55. Which sentence best states the purpose of the experiment and should remain as the opening?

- A) Sentence 2
- B) Sentence 1
- C) Sentence 4
- D) Sentence 7

56. Which revision of sentence 3 uses correct and effective subordination?

- A) Although one table was covered by the sail, the other remained in direct sunlight.
- B) Covering one table with the sail and the other remained in direct sunlight.
- C) One table was covered by the sail, the other remained in direct sunlight.
- D) One table covered by the sail; while the other was in direct sunlight.

57. Which revision of sentence 7 is most precise?

- A) The shaded table was kind of cooler in the readings.
- B) The table under the sail had temperatures and the other did too.
- C) The results were good for the shade sail on the tested days.
- D) On each clear-day trial, most readings at the shaded table were lower than those at the unshaded table.

58. Which revision of sentence 8 correctly uses parallel structure?

- A) The club presented its results to the principal and recommended testing longer across different seasons.
- B) The club presented its results to the principal and recommending a longer test in different seasons.
- C) The club was presenting its results and a longer test was recommended.
- D) The club presented its results, and the recommendation of a longer seasonal test.

59. Which sentence is written correctly?

- A) Neither of the two routes lead directly to the museum.
- B) Neither of the two routes leads directly to the museum.
- C) Neither of the two routes leading directly to the museum.
- D) Neither of the two routes have led direct to the museum.

60. Which sentence uses pronouns correctly?

- A) The coach gave the practice schedule to Elena and I.
- B) Me and Elena received the practice schedule from the coach.
- C) The coach gave Elena and me the practice schedule.
- D) The coach gave she and I the practice schedule.

61. Which sentence is clearest and most concise?

- A) Due to the fact that the bridge was closed, we took a route that was different.
- B) The bridge being closed was the reason why our route was changed by us.
- C) We took another route, and this was because of the closed condition of the bridge.
- D) Because the bridge was closed, we took another route.

62. Which numbered line contains an error in standard written US English?

Line 1: The collection of postcards was stored in a wooden box.

Line 2: Each card showed a landmark from a different city.

Line 3: The oldest cards were handled more careful than the others.

Line 4: Visitors could examine digital copies on a nearby screen.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

63. Which numbered line contains an error in standard written US English?

Line 1: Before the workshop began, every participant received safety glasses.

Line 2: The instructor demonstrated how to secure the pieces firmly.

Line 3: Several students asked questions about the final assembly.

Line 4: After the demonstration, everyone return to a separate worktable.

- A) Line 1
- B) Line 4
- C) Line 2
- D) Line 3

Revision Passage: A Reusable Container Return Station

[1] Our student council tested a return station for reusable lunch containers. [2] Students placed empty containers on a sloped rack beside the cafeteria exit. [3] The rack guided each container toward one of three labeled bins. [4] During the first lunch period, several students paused because the labels were printed too small. [5] We replaced them with larger labels and added a sample container above each bin. [6] After the change, the line moved more quickly. [7] The station collected 214 containers in one week. [8] Clear directions made the return process easier for students and cafeteria workers.

64. Which sentence would best follow sentence 3 to explain how the bins were used?

- A) The cafeteria serves lunch during three periods.
- B) Some containers were round, while others were square.
- C) One bin held cups, one held bowls, and one held divided trays.
- D) The council meets twice each month after school.

65. Which revision of sentence 4 is most effective?

- A) During the first lunch period, several students paused because the labels were too small to read quickly.
- B) During the first lunch period several students paused, the labels were printed too small.
- C) Several students, during the first lunch period, because the labels were printed too small.
- D) The labels printed too small, and several students during the first lunch period paused.

66. Which revision of sentence 6 is most precise?

- A) After that, things improved a lot.
- B) Then the whole situation became better for everyone.
- C) The change was nice, and the line did something different.
- D) After the larger labels were added, students sorted containers with fewer delays.

67. Which sentence best concludes the passage if sentence 8 is replaced?

- A) Reusable containers are sold in many sizes and colors.
- B) The trial showed that a return system works best when its categories are immediately clear.
- C) Some students brought lunches from home during the week.
- D) The cafeteria exit is also near the main staircase.

Revision Passage: Training the Morning Announcers

[1] Four students volunteered to deliver the school's morning announcements. [2] Each student practiced reading the same short script into a microphone. [3] The recordings were reviewed for pace, volume, and clear pronunciation. [4] One speaker rushed through dates, while another paused so often that the message sounded disconnected. [5] After listening, the students marked places to breathe and words that needed emphasis. [6] They recorded the script again, and every version was easier to understand. [7] The microphone stand was made of black metal. [8] The exercise showed that effective speaking requires both preparation and careful listening.

68. "Where should the following sentence be placed?

"The group also noted whether each speaker sounded confident rather than memorized."

- A) Before sentence 1
- B) After sentence 1
- C) After sentence 3
- D) After sentence 8

69. Which revision most effectively combines sentences 4 and 5?

- A) One speaker rushed through dates, another paused often, the students marked the script.
- B) After hearing that one speaker rushed and another paused too often, the students marked breathing points and words for emphasis.
- C) The students marked the script, one speaker rushed, and another paused.
- D) Rushing through dates and pausing often, the script was marked by the students.

70. Which sentence should be deleted because it does not support the passage's focus?

- A) Sentence 7
- B) Sentence 4
- C) Sentence 6
- D) Sentence 8

71. Which revision of sentence 2 uses standard written US English?

- A) Each student practice reading the same short script.
- B) Each student practiced reading the same short script into a microphone.
- C) Each student have practiced reading the same short script.
- D) Each student were practicing the same short script.

72. Which sentence uses the pronoun correctly?

- A) The advisor spoke with Maya and I after the meeting.
- B) The advisor spoke with Maya and myself after the meeting.
- C) The advisor spoke with Maya and me after the meeting.
- D) The advisor spoke with she and Maya after the meeting.

73. Which sentence is written correctly?

- A) Neither of the two signs is visible from the doorway.
- B) Neither of the two signs are visible from the doorway.
- C) Neither of the two sign is visible from the doorway.
- D) Neither of the two signs were visible from the doorway now.

74. Which line contains an error in standard written US English?

Line 1: The committee reviewed the three proposals.

Line 2: Each proposal included a cost estimate.

Line 3: The members discussed the options calmly.

Line 4: One of the plans require additional storage.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

75. Which line contains an error in standard written US English?

Line 1: The boxes arrived on Tuesday.

Line 2: Volunteers opened them carefully.

Line 3: The labels were bright and clearly.

Line 4: Every item was counted before storage.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

76. Which line contains an error in standard written US English?

Line 1: Before the doors opened, the guides checked their notes.

Line 2: Every guide carried a printed schedule.

Line 3: The visitors asked several useful questions.

Line 4: Later, the guides answers were collected.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

Revision Passage: A Student Podcast Booth

[1] Our media club proposed converting an unused storage room into a small podcast booth. [2] Before planning the space, club members surveyed students and teachers about how they might use it. [3] Most respondents wanted a quiet place for interviews, oral-history projects, and language practice. [4] The school's mascot is a falcon. [5] We measured the room and selected removable sound-absorbing panels that would not damage the walls. [6] The club also created a reservation calendar and simple instructions for operating the recorder. [7] A short training session will be required before anyone may reserve the booth independently. [8] With clear procedures and modest equipment, the unused room can become a practical resource for many classes.

77. Which numbered sentence is unrelated to the proposal for the podcast booth and should be removed?

- A) Sentence 2
- B) Sentence 4
- C) Sentence 6
- D) Sentence 8

78. Which revision of sentence 1 is most precise?

- A) Our club had an idea involving a room and podcasts.
- B) A storage room was something our media club thought about changing.
- C) Our media club proposed turning an unused storage room into a compact recording booth for student projects.
- D) There was an unused room, and the media club proposed it.

79. Which revision most effectively combines sentences 5 and 6?

- A) After selecting removable sound-absorbing panels for the measured room, the club created a reservation calendar and operating instructions.
- B) The room was measured, the panels were removable, and also there was a calendar with instructions.
- C) Selecting panels that would not damage the walls, a calendar and instructions were created by the club.
- D) The club measured the room because a reservation calendar operated the recorder.

80. Which sentence would provide the most relevant support after sentence 6?

- A) Some students listen to podcasts while traveling to school.
- B) The storage room once contained extra stage curtains.
- C) Professional recording studios may use expensive mixing equipment.
- D) The calendar will limit each reservation to forty minutes so that more groups can use the booth.

Revision Passage: Comparing Lunchbox Liners

[1] The science club wanted to compare two materials that might help keep lunches cool. [2] Members lined one identical container with felt and another with reflective foam. [3] They placed the same number of ice cubes in each container and recorded the temperature every fifteen minutes. [4] During the first trial, however, the felt-lined container sat closer to a sunny window. [5] Because the containers had not received equal amounts of sunlight, the results could not provide a fair comparison. [6] The club repeated the test with both containers on the same shaded table. [7] In the repeated trials, the reflective-foam container remained cooler for a longer period. [8] The project showed that a useful experiment depends on controlling conditions as carefully as measuring results.

81. Which sentence would provide the strongest introduction to the experiment described in the passage?

- A) Lunch containers are sold in many shapes and colors.
- B) The science club meets every Thursday after school.
- C) A fair comparison requires researchers to keep most conditions the same while changing only the factor being tested.
- D) Reflective foam is usually lighter than felt.

82. Which transition best begins sentence 5?

- A) For example,
- B) As a result,
- C) Meanwhile,
- D) In addition,

83. Which version of sentence 6 states the same idea most directly and concisely?

- A) The club did the test again, and this time the placement was a placement that was shaded.
- B) On a shaded table was where the club repeated the test with both containers together.
- C) The test was repeated by the club with both of the containers being on the same table in shade.
- D) The club repeated the test with both containers on the same shaded table.

84. Which concluding sentence most effectively reinforces the passage's main point?

- A) By correcting the unequal placement, the club produced evidence that could support a dependable comparison.
- B) The containers were returned to the supply cabinet after the experiment.
- C) Many students bring different foods for lunch each day.
- D) The club may test the materials in containers of different colors.

85. Which option follows standard written US English?

- A) Each of the volunteers have received a schedule.
- B) Each of the volunteers has received a schedule.
- C) Each of the volunteers have receives a schedule.
- D) Each of the volunteers has receive a schedule.

86. In which sentence are all pronouns in the correct case?

- A) The librarian gave the research forms to Maya and I.
- B) Him and Elena presented the project first.
- C) The coach asked she and Marcus to remain after practice.
- D) The guide showed Carlos and me where the trail began.

87. Which numbered line contains an error in standard written US English?

Line 1: The orchestra members arrived before noon.

Line 2: They placed their instrument cases along the back wall.

Line 3: Neither the conductor nor the musicians was ready to begin.

Line 4: After a brief sound check, the rehearsal started.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

88. Which numbered line contains an error in standard written US English?

Line 1: The two student's model was displayed near the entrance.

Line 2: Its moving parts were powered by a small motor.

Line 3: Visitors watched as the gears turned slowly.

Line 4: A label explained how the mechanism worked.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

Revision Passage: Improving Locker-Number Visibility

[1] Students reported that locker numbers were difficult to read from the center of the hallway. [2] Our design class photographed the lockers from several distances. [3] We compared the existing silver numbers with paper samples in dark blue, black, and white. [4] The black samples produced the strongest contrast against the pale doors. [5] We also tested two number sizes during a passing period. [6] More students identified the larger numbers correctly while walking. [7] The hallway contains eighty-four lockers. [8] We recommended larger black numbers placed at the same height on every door.

89. Which sentence would best follow sentence 2?

- A) Some students prefer lockers near the stairwell.
- B) The photographs showed that glare and low contrast hid many numbers.
- C) The class camera had a rechargeable battery.
- D) The hallway floor was polished every Friday.

90. Which revision of sentence 5 is most effective?

- A) During a passing period, we also tested two number sizes.
- B) Two number sizes during a passing period, and we tested them.
- C) During a passing period two number sizes were, they were tested.
- D) Testing two number sizes during a passing period, the students.

91. Which revision of sentence 6 is clearest?

- A) While walking through the hall, students correctly read the larger numbers more often.
- B) The larger numbers were better and got more correct.
- C) Walking, the numbers were identified by more students correctly.
- D) More students did it right when the numbers got bigger.

92. Which sentence should be deleted because it is not needed to develop the passage?

- A) Sentence 2
- B) Sentence 4
- C) Sentence 7
- D) Sentence 8

Revision Passage: Building a Miniature Wind Tunnel

[1] Our science group built a small wind tunnel from a clear storage box and a quiet fan. [2] A cardboard grid at the entrance straightened the moving air. [3] We suspended paper shapes from a thin rod inside the box. [4] Each shape was tested at the same fan setting for thirty seconds. [5] The narrow triangle moved less than the wide rectangle. [6] Because the shapes had different areas, we could not claim that outline alone caused the difference. [7] We plan to repeat the test with shapes cut to equal areas. [8] The first trial helped us design a fairer second experiment.

93. Which revision most effectively combines sentences 2 and 3?

- A) After a cardboard grid straightened the moving air, we suspended paper shapes from a thin rod inside the box.
- B) A cardboard grid straightened the air, the paper shapes were suspended from a rod.
- C) Straightening the air, the paper shapes suspended from a rod.
- D) The grid and the shapes, which were in the box and straightened.

94. Which revision of sentence 5 is most precise?

- A) The triangle did better than the rectangle.
- B) One shape moved in a different way from another shape.
- C) At the same fan setting, the narrow triangle swung through a smaller distance than the wide rectangle.
- D) The shapes acted like shapes in moving air.

95. Which revision of sentence 7 uses standard written US English?

- A) We plans to repeat the test with equal areas.
- B) We plan to repeat the test with shapes cut to equal areas.
- C) We planning to repeat the test with shapes of equal area.
- D) We plan repeating the test with equal areas yesterday.

96. Which sentence is written correctly?

- A) The boxes that was delivered today are in the office.
- B) The boxes that were delivered today are in the office.
- C) The boxes who were delivered today is in the office.
- D) The boxes delivered today was in the office.

97. Which sentence uses the modifier correctly?

- A) Walking into the lab, the experiment was already running.
- B) Covered in labels, Maya organized the storage shelf.
- C) Walking into the lab, Maya noticed that the experiment was already running.
- D) After reading the directions, the machine was started.

98. Which line contains an error in standard written US English?

Line 1: The model rested on a wooden base.

Line 2: Two wires connected it to the battery.

Line 3: The students tested the switch twice.

Line 4: Each of the lights were visible from the doorway.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

99. Which line contains an error in standard written US English?

Line 1: The guide spoke clearly to the group.

Line 2: Everyone followed her into the workshop.

Line 3: She explained the safety rules first.

Line 4: Then she demonstrated the tool careful.

A) Line 1

B) Line 2

C) Line 3

D) Line 4

100. Which line contains an error in standard written US English?

Line 1: The team finished its first draft.

Line 2: Two paragraphs needed additional evidence.

Line 3: The writers revised those sections.

Line 4: The editors comments were added in blue.

A) Line 1

B) Line 2

C) Line 3

D) Line 4

Section 3: Mathematics

50 questions | Recommended time: 45 minutes

101. A recipe uses $2\frac{1}{4}$ cups of oats for one batch. How many cups are needed for 3 batches?

- A) $6\frac{3}{4}$
- B) $6\frac{1}{4}$
- C) $7\frac{1}{2}$
- D) $8\frac{1}{4}$

102. What is the value of $18 - 3 \times (4 + 2)$?

- A) -90
- B) 0
- C) 36
- D) 90

103. At a fundraiser, 36 of 90 tickets were sold before noon. What percent of the tickets were sold?

- A) 25%
- B) 36%
- C) 40%
- D) 54%

104. Solve $5(x - 2) = 35$.

- A) 5
- B) 7
- C) 8
- D) 9

105. The first four terms of a sequence are 4, 9, 14, 19. Which expression gives the n th term?

- A) $5n - 1$
- B) $4n + 1$
- C) $5n + 4$
- D) $n + 5$

106. A streaming service charges a fixed fee of \$6 plus \$2 for each movie rented. Which equation gives the total cost c for renting m movies?

- A) $c = 6m + 2$
- B) $c = 2(m + 6)$
- C) $c = 2m + 6$
- D) $c = 8m$

107. The numbers of books read by five students are 3, 5, 5, 7, and 10. What is the mean?

- A) 5
- B) 5.5
- C) 7
- D) 6

108. A bag contains 4 red, 3 blue, and 5 green tiles. One tile is chosen, replaced, and then another is chosen. What is the probability that both tiles are blue?

- A) $\frac{1}{12}$
- B) $\frac{1}{16}$
- C) $\frac{1}{8}$
- D) $\frac{1}{4}$

109. Two angles are supplementary. One angle measures 118° . What is the measure of the other angle?

- A) 62°
- B) 72°
- C) 118°
- D) 242°

110. A square has an area of 81 cm^2 . What is its perimeter?

- A) 18 cm
- B) 36 cm
- C) 40 cm
- D) 81 cm

111. A cyclist travels 18 miles in 1.5 hours at a constant rate. What is the cyclist's average speed?

- A) 9 miles per hour
- B) 10 miles per hour
- C) 12 miles per hour
- D) 27 miles per hour

112. A model uses a scale of 1 inch = 6 feet. A wall is 4.5 inches long on the model. How long is the actual wall?

- A) 10.5 feet
- B) 18 feet
- C) 24 feet
- D) 27 feet

113. What is the value of $|-18| - |7 - 12|$?

- A) 5
- B) 11
- C) 12
- D) 13

114. What is the value of $\frac{5}{6} - \frac{1}{4} + \frac{1}{3}$?

- A) $\frac{7}{12}$
- B) $\frac{11}{12}$
- C) $\frac{13}{12}$
- D) $\frac{17}{12}$

115. How many positive factors does 36 have?

- A) 9
- B) 8
- C) 10
- D) 12

116. A line passes through the points (2, 5) and (6, 17). What is its rate of change?

- A) 2
- B) 2.5
- C) 3
- D) 4

117. Which expression is equivalent to $3(2x - 5) + 4x$?

- A) $6x - 1$
- B) $7x - 15$
- C) $10x + 15$
- D) $10x - 15$

118. A frequency list contains one 2, three 3s, five 4s, and one 5. What is the mean?

- A) 3.6
- B) 3.8
- C) 4
- D) 4.2

119. One letter is selected from the word MATH, and one number is selected from 1, 2, and 3. What is the probability of selecting a vowel and an even number?

- A) $\frac{1}{6}$
- B) $\frac{1}{12}$
- C) $\frac{1}{4}$
- D) $\frac{1}{3}$

120. The ordered pairs (1, 12), (2, 18), (3, 25), and (4, 31) compare hours practiced with points earned. Which conclusion is best supported?

- A) Points decrease as practice time increases.
- B) Practice time and points have no relationship.
- C) Every additional hour always adds exactly 6 points.
- D) The data show a positive association between practice time and points.

121. Three vertices of a rectangle are (1, 2), (1, 7), and (6, 2). Which point is the fourth vertex?

- A) (2, 6)
- B) (5, 7)
- C) (6, 7)
- D) (7, 6)

122. A 90° central angle cuts a circle into a sector. What fraction of the circle is the sector?

- A) $\frac{1}{2}$
- B) $\frac{1}{4}$
- C) $\frac{1}{3}$
- D) $\frac{3}{4}$

123. A cylinder has a radius of 3 centimeters and a height of 5 centimeters. Using 3.14 for π , what is its volume?

- A) 141.3 cubic centimeters
- B) 94.2 cubic centimeters
- C) 47.1 cubic centimeters
- D) 282.6 cubic centimeters

124. A faucet fills 9 liters in 6 minutes at a constant rate. How many milliliters does it fill in 4 minutes?

- A) 4,500 milliliters
- B) 5,400 milliliters
- C) 6,000 milliliters
- D) 7,500 milliliters

125. Use 1 square meter = 10,000 square centimeters. How many square centimeters are equal to 2.5 square meters?

- A) 250
- B) 2,500
- C) 20,000
- D) 25,000

126. A recipe uses $\frac{3}{4}$ cup of rice for 6 servings. At the same rate, how many cups of rice are needed for 14 servings?

- A) $1\frac{1}{4}$ cups
- B) $1\frac{1}{2}$ cups
- C) $1\frac{2}{3}$ cups
- D) $1\frac{3}{4}$ cups

127. A jacket originally costs \$80. Its price is increased by 15%. What is the new price?

- A) \$92
- B) \$88
- C) \$95
- D) \$104

128. Evaluate $-4 + 13 - 9$.

- A) -18
- B) -8
- C) 0
- D) 8

129. Solve $4x + 7 = 31$.

- A) 5
- B) 6
- C) 7
- D) 8

130. What is the next term in the sequence 2, 6, 18, 54, ...?

- A) 162
- B) 108
- C) 72
- D) 58

131. For the data set 6, 8, 8, 10, 13, how much greater is the mean than the median?

- A) 0
- B) $\frac{1}{2}$
- C) 2
- D) 1

132. A bag contains 3 red tiles, 4 blue tiles, and 5 yellow tiles. If one tile is selected at random, what is the probability that it is not blue?

- A) $\frac{1}{3}$
- B) $\frac{2}{3}$
- C) $\frac{3}{4}$
- D) $\frac{5}{12}$

133. An exterior angle of a triangle measures 124° . One of its remote interior angles measures 47° . What is the measure of the other remote interior angle?

- A) 67°
- B) 71°
- C) 77°
- D) 81°

134. An L-shaped floor can be viewed as a 10-foot by 8-foot rectangle with a 4-foot by 3-foot rectangle removed. What is the area of the floor?

- A) 56 square feet
- B) 62 square feet
- C) 72 square feet
- D) 68 square feet

135. A rectangular prism is 7 centimeters long, 5 centimeters wide, and 4 centimeters high. What is its volume?

- A) 140 cm^3
- B) 80 cm^3
- C) 70 cm^3
- D) 16 cm^3

136. How many centimeters are equal to 2.75 meters?

- A) 27.5
- B) 250
- C) 275
- D) 2,750

137. Points P(-2, 1) and Q(4, 1) lie on a coordinate plane. What is the distance between P and Q?

- A) 4 units
- B) 6 units
- C) 7 units
- D) 8 units

138. Which fraction is exactly halfway between $\frac{2}{5}$ and $\frac{3}{5}$?

- A) $\frac{3}{8}$
- B) $\frac{9}{20}$
- C) $\frac{1}{2}$
- D) $\frac{11}{20}$

139. Which number is equal to $2^3 \times 3^2$?

- A) 36
- B) 48
- C) 64
- D) 72

140. What is the smallest whole number that must be added to 587 to make the result divisible by 9?

- A) 7
- B) 5
- C) 4
- D) 2

141. Which ordered pair satisfies both $y = -2x + 5$ and $x + y = 2$?

- A) (1, 1)
- B) (3, -1)
- C) (-1, 3)
- D) (2, 0)

142. A line has a slope of -3 and passes through (2, 1). What is the y-coordinate when $x = 5$?

- A) -14
- B) -10
- C) -8
- D) 8

143. The data set is 4, 5, 6, 7, 8. If 30 is added, which measure changes more: the mean or the median?

- A) The mean changes more.
- B) The median changes more.
- C) They change by the same amount.
- D) Neither measure changes.

144. A whole number from 1 through 20 is selected at random. What is the probability that it is divisible by 3 or by 5?

- A) $\frac{7}{20}$
- B) $\frac{2}{5}$
- C) $\frac{1}{2}$
- D) $\frac{9}{20}$

145. A runner completes 3 laps in 12 minutes each and 2 laps in 18 minutes each. What is the mean time per lap?

- A) 13.2 minutes
- B) 14 minutes
- C) 14.4 minutes
- D) 15 minutes

146. A parallelogram has an area of 84 square centimeters and a base of 12 centimeters. What is its height?

- A) 5 centimeters
- B) 7 centimeters
- C) 8 centimeters
- D) 14 centimeters

147. A rectangle has vertices $(-2, -1)$, $(-2, 4)$, $(5, 4)$, and $(5, -1)$. What is its area?

- A) 24 square units
- B) 28 square units
- C) 30 square units
- D) 35 square units

148. The point $(4, -3)$ is reflected across the y-axis. What are the coordinates of its image?

- A) $(-4, -3)$
- B) $(4, 3)$
- C) $(-3, 4)$
- D) $(3, -4)$

149. A 3.5-pound bag of apples costs \$8.75. What is the cost per pound?

- A) \$2.25
- B) \$2.50
- C) \$2.75
- D) \$3.00

150. Use 1 cubic meter = 1,000 liters. A container has a volume of 0.012 cubic meter. How many liters does it hold?

- A) 1.2 liters
- B) 10 liters
- C) 12 liters
- D) 120 liters

Section 4: Ability

50 questions | Recommended time: 35 minutes

151. Complete the 2×2 matrix.

Top-left: ○

Top-right: ○○○

Bottom-left: ■

Bottom-right: ?

Across each row, two copies of the same shape are added. Moving down changes an open shape to a filled square.

- A) ■■
- B) □□□
- C) ●●●
- D) ■■■■

152. Complete the 3×3 matrix.

Row 1: ↑ | → | ↓

Row 2: → | ↓ | ←

Row 3: ↓ | ← | ?

Each step to the right or downward rotates the arrow 90° clockwise.

- A) ↑
- B) →
- C) ↓
- D) ←

153. Complete the 2×2 matrix.

Top-left: ○ inside □

Top-right: □ inside ○

Bottom-left: △ inside ◇

Bottom-right: ?

Across each row, the inner and outer shapes exchange positions.

- A) △ inside ◇
- B) ◇ inside △
- C) ▲ inside ◆
- D) ◇ beside △

154. Complete the 3×3 matrix.

Row 1: ○ | □ | ○□

Row 2: ● | ■ | ●■

Row 3: △ | ▲ | ?

In each row, the third cell places the first two symbols together in their original order.

- A) ▲△
- B) △■
- C) △▲
- D) ▲▲

155. Complete the 2×2 matrix.

Top-left: open ○ **above** open □

Top-right: filled ● **above** filled ■

Bottom-left: open △ **above** open ◇

Bottom-right: ?

Across each row, positions stay fixed while both shapes change from open to filled.

- A) filled ▲ **above** filled ◆
- B) open △ **above** open ◇
- C) filled ◆ **above** filled ▲
- D) filled ▲ **above** open ◇

156. A rectangular paper is folded in half from right to left. One hole is punched near the lower-left corner of the folded rectangle, away from all edges. After unfolding, where are the holes?

- A) Near both lower corners, symmetric across the vertical centerline
- B) Near both upper corners
- C) One near the lower-left corner only
- D) Four holes, one in each quadrant

157. A square paper is folded from top to bottom and then from right to left. One hole is punched near the upper-right corner of the final folded square, away from all edges. What appears after unfolding?

- A) Two holes near the top edge
- B) Four holes, one in each quadrant, symmetric across both centerlines
- C) Four holes along the left edge
- D) Eight holes around the center

158. A square paper is folded once along the diagonal from upper-right to lower-left. A small triangular notch is cut from the folded diagonal, away from its endpoints. What appears after unfolding?

- A) Two separate triangular notches on opposite outer edges
- B) One triangular notch at the upper-right corner
- C) A symmetric diamond-shaped opening centered on the diagonal
- D) Four triangular openings around the center

159. A square paper is folded in half from left to right. The folded rectangle is then folded from bottom to top. A hole is punched near the center of the final folded square, away from all edges. After unfolding, how many holes appear?

- A) 1
- B) 2
- C) 8
- D) 4

160. The arrangements ○|○, □|□, and △|△ share a rule. Which choice follows the same rule?

- A) ◇|◆
- B) ☆|☆
- C) ●|○
- D) +|×

161. The figures ○●, □■, and △▲ share a rule. Which choice follows the same rule?

- A) ◆◇
- B) ☆○
- C) ◇◆
- D) ■□

162. The arrangements “○ above □,” “△ above ◇,” and “☆ above +” share a rule. Which choice follows the same rule?

- A) ■ above ◆
- B) + above ☆
- C) ● beside □
- D) ◇ above ○

163. Complete the 2 × 2 matrix.

Top-left: ↑ ○

Top-right: → ●

Bottom-left: ← □

Bottom-right: ?

Moving right rotates the arrow 90° clockwise and changes the open shape to filled.

- A) ↑ ■
- B) ↓ □
- C) ↓ ■
- D) ← ■

164. Complete the 2 × 3 matrix.

Row 1: ○○ | ○○○ | ●

Row 2: □□ | □□□□ | ?

In each row, subtract the first count from the second count, then fill the remaining shapes.

- A) □□
- B) ■■■■
- C) ■■
- D) ■

165. Complete the 3 × 3 matrix.

Row 1: ○ | □□ | △△△

Row 2: □□ | △△△ | ○

Row 3: △△△ | ○ | ?

Each row shifts the three cells one position left.

- A) ○○
- B) △△△
- C) □
- D) □□

166. Complete the 2×2 matrix.

Top-left: Top-left ○

Top-right: Top-right ●

Bottom-left: Bottom-left □

Bottom-right: ?

Moving right shifts the symbol horizontally and fills it.

- A) Top-left ■
- B) Bottom-right ■
- C) Bottom-right □
- D) Top-right ■

167. A square sheet is folded from bottom to top and then from right to left. One hole is punched near the lower-right corner of the final folded square, away from all edges and folds. What appears after unfolding?

- A) Two holes in a vertical pair
- B) Two holes in a horizontal pair
- C) Three holes forming an L
- D) Four holes in a 2×2 arrangement

168. A rectangular sheet is divided into four equal vertical panels and folded accordion-style into one panel. One interior hole is punched through the stack. What appears after unfolding?

- A) Four holes in one horizontal row
- B) Four holes in one vertical column
- C) Eight holes in two rows
- D) Two holes near the center

169. A square sheet is folded once along the diagonal from the upper-left corner to the lower-right corner. One hole is punched near the folded triangle's upper-left side, away from the fold. What appears after unfolding?

- A) One hole on the diagonal
- B) Two holes reflected across the diagonal
- C) Four holes near the corners
- D) Two holes reflected across a vertical line

170. A square sheet is folded from left to right. A semicircular notch is cut at the midpoint of the folded edge created by the fold. What appears after unfolding?

- A) Two notches on the outer edges
- B) Four holes around the center
- C) One circular hole centered on the vertical midline
- D) One semicircular notch on the left edge

171. The following arrangements share the same rule:

● ○ ●

■ □ ■

▲ △ ▲

Which choice follows the same rule?

- A) ◆ ◇ ◆
- B) ◇ ◆ ◇
- C) ◆ ◆ ◇
- D) ◇ ◇ ◆

172. The following groups share the same rule:

○ | ○

□ - □

△ / △

Which choice follows the same rule?

A) ◇ ◆ ◇

B) ◇ ◇ |

C) ◆ \ ◆

D) ◇ \ ◇

173. The following arrangements share the same rule:

● ○ ○

■ □ □

▲ △ △

Which choice follows the same rule?

A) ◇ ◆ ◆

B) ◆ ◇ ◆

C) ◆ ◇ ◇

D) ◇ ◇ ◆

174. The following groups share the same rule:

↑ → ↓

→ ↓ ←

↓ ← ↑

Which choice follows the same rule?

A) ↑ ↓ ←

B) ← ↑ →

C) ← → ↑

D) → ↑ ←

175. Complete the 2 × 2 matrix.

Top-left: ○

Top-right: ●

Bottom-left: □

Bottom-right: ?

Across each row, the shape changes from open to filled. Down each column, a circle changes to a square.

A) ■

B) □

C) ▲

D) ●

176. Complete the 3 × 3 matrix.

Row 1: ○ | □ | △

Row 2: △ | ○ | □

Row 3: □ | △ | ?

Each row and column contains one circle, one square, and one triangle.

A) □

B) △

C) ○

D) ◇

177. Complete the 2×2 matrix.

Top-left: $\uparrow \circ$

Top-right: $\rightarrow \circ \circ$

Bottom-left: $\downarrow \square$

Bottom-right: ?

Across each row, the arrow rotates 90° clockwise and one shape is added. Down each column, the arrow rotates 180° and circles change to squares.

A) $\rightarrow \square \square$

B) $\leftarrow \square \square$

C) $\leftarrow \circ \circ$

D) $\uparrow \square$

178. Open shapes: $\circ \triangle \square \diamond$

Filled shapes: $\bullet \blacktriangle \blacksquare \blacklozenge$

Complete the 2×2 matrix.

Top-left: $\circ$ inside $\square$

Top-right: $\bullet$ inside $\square$

Bottom-left: $\triangle$ inside $\diamond$

Bottom-right: ?

Across each row, only the inner shape changes from open to filled.

A) $\triangle$ inside $\blacklozenge$

B) $\blacktriangle$ inside $\blacklozenge$

C) $\bullet$ inside $\diamond$

D) $\blacktriangle$ inside $\diamond$

179. Complete the 3×3 matrix.

Row 1: $\uparrow \mid \leftarrow \mid \downarrow$

Row 2: $\leftarrow \mid \downarrow \mid \rightarrow$

Row 3: $\downarrow \mid \rightarrow \mid ?$

Across each row and down each column, the arrow rotates 90° counterclockwise.

A) $\leftarrow$

B) $\downarrow$

C) $\uparrow$

D) $\rightarrow$

180. A square sheet is folded in half from left to right. One hole is punched near the upper-right corner of the folded rectangle, away from the fold. What appears when the sheet is unfolded?

A) Two holes near the upper-left and upper-right corners, mirrored across the vertical centerline

B) Two holes near the lower-left and lower-right corners

C) Four holes, one near each corner

D) One hole on the vertical centerline

181. A square sheet is folded from top to bottom and then from left to right. One hole is punched near the lower-right corner of the final folded square, away from both folds. What appears after complete unfolding?

A) One hole near the lower-right corner

B) Two holes mirrored across the vertical centerline

C) Two holes mirrored across the horizontal centerline

D) Four holes, one in each quadrant, symmetric across both centerlines

182. A square sheet is folded once along the diagonal from upper-left to lower-right. A small triangular notch is cut on the folded diagonal near the lower-right end, away from the corner. What appears when the sheet is unfolded?

- A) Two separate triangular cuts on opposite edges
- B) One symmetric diamond-shaped opening on the diagonal
- C) Four triangular openings around the center
- D) One semicircular opening on the diagonal

183. A square sheet is folded in half from top to bottom. A semicircular notch is cut at the midpoint of the folded sheet's left outer edge, away from the horizontal fold. What appears after unfolding?

- A) One circular hole at the center
- B) Two notches on the top and bottom edges
- C) Two matching semicircular notches on the left edge, mirrored across the horizontal centerline
- D) Four notches, one on each edge

184. The three arrangements below share the same rule.

Arrangement 1: ○ ● ○

Arrangement 2: □ ■ □

Arrangement 3: △ ▲ △

Which choice follows the same rule?

- A) ☆ ☆ ★
- B) ◇ ◆ ◆
- C) + × + ×
- D) ☆ ★ ☆

185. The three sequences below share the same rule.

Sequence 1: ↑ → ↓

Sequence 2: → ↓ ←

Sequence 3: ↓ ← ↑

Which choice follows the same rule?

- A) ← ↑ →
- B) ← → ↑
- C) ↑ ← ↓
- D) → ↑ ↓

186. Open shapes: ○ △ □ ◇

Filled shapes: ● ▲ ■ ◆

The three figures below share the same rule.

Figure 1: ○ inside □; ● outside □

Figure 2: △ inside ◇; ▲ outside ◇

Figure 3: □ inside ○; ■ outside ○

Which choice follows the same rule?

- A) ○ inside □; ■ outside □
- B) ◇ inside △; ◆ outside △
- C) ▲ inside ◇; △ outside ◇
- D) □ inside ○; ◆ outside ○

187. Study the repeating structure in each example.

Example 1: ○ □ ○ □

Example 2: △ ◇ △ ◇

Example 3: ☆ + ☆ +

Which option uses the same alternating pattern?

A) ○ ○ □ □

B) △ ◇ ◇ △

C) ■ ● ■ ●

D) ☆ + + ☆

188. Complete the 2×2 matrix.

Top-left: ○ →

Top-right: ● ↓

Bottom-left: □ ↑

Bottom-right: ?

Moving right fills the shape and rotates the arrow 90° clockwise.

A) □ →

B) ■ ←

C) □ ↓

D) ■ →

189. Complete the 2×3 matrix.

Row 1: ○ ○ ○ | ○ | ● ●

Row 2: △ △ △ △ | △ △ △ | ?

In each row, subtract the second count from the first, then fill the result.

A) ▲

B) △

C) ▲ ▲

D) △ △

190. Complete the 3×3 matrix.

Row 1: ○ | □ | △

Row 2: □ | △ | ○

Row 3: △ | ○ | ?

Each row shifts the symbols one position left.

A) ○

B) △

C) □

D) ◇

191. Complete the 2×2 matrix.

Top-left: Left ○; Right ●

Top-right: Top ○; Bottom ●

Bottom-left: Left □; Right ■

Bottom-right: ?

Moving right rotates the arrangement 90° clockwise.

A) Top ■; Bottom □

B) Top □; Bottom ■

C) Left ■; Right □

D) Bottom □; Top ■

192. A rectangular sheet is folded from left to right, then from bottom to top. A hole is punched near the upper-left corner of the final packet, away from edges and folds. What appears after unfolding?

- A) Two holes in a horizontal row
- B) Two holes in a vertical column
- C) Three holes forming a triangle
- D) Four holes in a 2×2 arrangement

193. A square sheet is folded along the diagonal from lower-left to upper-right. A triangular notch is cut from the folded packet's outer bottom edge, away from the fold. What appears after unfolding?

- A) One notch on the diagonal
- B) Two matching notches reflected across the diagonal
- C) Four notches, one on each edge
- D) One diamond-shaped hole at the center

194. A long rectangle is divided into three equal horizontal strips. The top strip is folded down onto the middle, and the bottom strip is folded up onto the middle. One hole is punched near the left side of the final stack. What appears after unfolding?

- A) Three holes in a vertical column near the left side
- B) Three holes in a horizontal row
- C) Six holes in two columns
- D) Two holes near the left edge

195. A square sheet is folded from top to bottom. A semicircular notch is cut from the midpoint of the folded edge created by the fold. What appears after unfolding?

- A) Two notches on the top and bottom edges
- B) Two holes beside the center
- C) Four semicircular notches
- D) One circular hole centered on the horizontal midline

196. The following groups share the same rule:

○ ● ○
 □ ■ □
 △ ▲ △

Which choice follows the same rule?

- A) ◆ ◇ ◆
- B) ◇ ◆ ◇
- C) ◇ ◇ ◆
- D) ◆ ◆ ◇

197. The following arrangements share the same rule:

↑ ○ ○
 → □ □
 ↓ △ △

Which choice follows the same rule?

- A) ← ◇ ◇
- B) ← ◆ ◆
- C) ↑ ◇ ◇
- D) ← ◇ ◆

198. The following groups share the same rule:

○ | ●

□ - ■

△ / ▲

Which choice follows the same rule?

A) ◇ ◆ \

B) ◆ \ ◇

C) ◇ \ ◆

D) ◇ ◆ |

199. The following arrangements share the same rule:

↑ → ←

→ ↓ ↑

↓ ← →

Which choice follows the same rule?

A) ← ↓ ↑

B) ← → ↓

C) ← ↑ ↓

D) ↑ ← ↓

200. The following figures share the same rule:

Open circle ○ containing ■

Open square □ containing ▲

Open diamond ◇ containing ●

Which choice follows the same rule?

A) Filled triangle ▲ containing ◆

B) Open triangle △ containing ◇

C) Filled triangle ▲ containing ○

D) Open triangle △ containing ◆

Answer Key

Question numbers continue across all five sections.

Reading

1. A	2. B	3. C	4. D	5. A	6. B	7. C	8. D
9. A	10. B	11. C	12. D	13. A	14. B	15. D	16. A
17. C	18. A	19. C	20. D	21. B	22. C	23. A	24. B
25. D	26. C	27. A	28. D	29. B	30. B	31. D	32. A
33. C	34. D	35. B	36. C	37. A	38. D	39. C	40. A
41. D	42. B	43. B	44. D	45. A	46. C	47. D	48. B
49. C	50. A						

Written Expression

51. B	52. C	53. A	54. D	55. B	56. A	57. D	58. A
59. B	60. C	61. D	62. C	63. B	64. C	65. A	66. D
67. B	68. C	69. B	70. A	71. B	72. C	73. A	74. D
75. C	76. D	77. B	78. C	79. A	80. D	81. C	82. B
83. D	84. A	85. B	86. D	87. C	88. A	89. B	90. A
91. A	92. C	93. A	94. C	95. B	96. B	97. C	98. D
99. D	100. D						

Mathematics

101. A	102. B	103. C	104. D	105. A	106. C	107. D	108. B
109. A	110. B	111. C	112. D	113. D	114. B	115. A	116. C
117. D	118. A	119. B	120. D	121. C	122. B	123. A	124. C
125. D	126. D	127. A	128. C	129. B	130. A	131. D	132. B
133. C	134. D	135. A	136. C	137. B	138. C	139. D	140. A
141. B	142. C	143. A	144. D	145. C	146. B	147. D	148. A
149. B	150. C						

Ability

151. D	152. A	153. B	154. C	155. A	156. A	157. B	158. C
159. D	160. B	161. C	162. D	163. A	164. C	165. D	166. B
167. D	168. A	169. B	170. C	171. A	172. D	173. C	174. B
175. A	176. C	177. B	178. D	179. C	180. A	181. D	182. B
183. C	184. D	185. A	186. B	187. C	188. D	189. A	190. C
191. B	192. D	193. B	194. A	195. D	196. B	197. A	198. C

199. C	200. D						
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Detailed Explanations

Reading

1. TACHS-R4-001 - Answer A: Understanding patterns and listening carefully help Jonah handle an unexpected problem. Jonah uses musical structure and attentive listening to perform successfully without the missing page.
2. TACHS-R4-002 - Answer B: Inside another piece of music
Backstage, Jonah finds the missing page tucked inside another musical piece.
3. TACHS-R4-003 - Answer C: He adjusts his approach instead of depending only on memorized movements. Jonah attends to the unfamiliar keys and listens to Elise rather than relying solely on practiced motion.
4. TACHS-R4-004 - Answer D: It provides Jonah with a way to reconstruct the measures he cannot remember. Recognizing that the measures echo an earlier phrase lets Jonah determine how the missing portion should sound.
5. TACHS-R4-005 - Answer A: Warm water is less dense than cold water.
The passage directly identifies the density difference as the reason warm water stays above cold water.
6. TACHS-R4-006 - Answer B: spreads into new locations
Turnover moves oxygen downward and nutrients upward, spreading them to different parts of the lake.
7. TACHS-R4-007 - Answer C: To clarify that a visible change can be part of a beneficial natural process
The author contrasts temporary cloudiness with turnover's benefits to prevent readers from assuming that cloudiness means pollution.
8. TACHS-R4-008 - Answer D: Surface water remaining much warmer than deep water
A strong temperature and density difference would continue separating the layers and prevent full mixing.
9. TACHS-R4-009 - Answer A: Effective maps select and simplify information according to a specific purpose.
The passage explains that mapmakers choose and generalize details to meet users' particular needs.
10. TACHS-R4-010 - Answer B: present in a simplified form
The examples of simplified coastlines and combined minor roads define how cartographers generalize.
11. TACHS-R4-011 - Answer C: Station order and transfer points
The second paragraph explicitly states that subway maps emphasize station sequence and transfers.
12. TACHS-R4-012 - Answer D: A map should be judged partly by how well it serves its intended users.
The passage says the best map presents the relationships its intended users need.
13. TACHS-R4-013 - Answer A: explain why mapmaking requires purposeful selection
The author describes how purpose and scale guide decisions about including, omitting, and simplifying information.
14. TACHS-R4-014 - Answer B: A mirror beside the frame
The mirror shows how the reversed letter arrangement will appear when printed.
15. TACHS-R4-015 - Answer D: Each proof reveals a problem that can still be corrected.
The proofs expose reversed text, uneven spacing, and margin problems before the final printing.
16. TACHS-R4-016 - Answer A: A polished result often depends on careful testing and correction.
Leah produces a simple-looking final poster by checking and correcting several hidden details.
17. TACHS-R4-017 - Answer C: a trial copy made for checking.
In printing, a proof is a test copy examined before the final version is produced.
18. TACHS-R4-018 - Answer A: Divide a strong air stream into smaller currents
The passage states that the projections split a strong stream of air into smaller currents.
19. TACHS-R4-019 - Answer C: To explain how biological principles can inspire quieter technology
The examples show how engineers adapt noise-reducing principles from owl feathers to machines.
20. TACHS-R4-020 - Answer D: It explains several feather features, their effects, and possible applications.
The passage moves from feather structures to their effects, trade-offs, and engineering uses.
21. TACHS-R4-021 - Answer B: advantages gained with accompanying disadvantages.
The quiet-flight features provide benefits but also reduce performance in some conditions.
22. TACHS-R4-022 - Answer C: By moving a pointer or lighting a floor lamp
The passage describes moving pointers and lamps activated as the car passed each floor.
23. TACHS-R4-023 - Answer A: The elevator could be farther away and harder to track without information.
More floors increased the possible distance and waiting uncertainty, making location information more valuable.

24. TACHS-R4-024 - Answer B: To show that old and new indicators address the same need for information
The final paragraph connects modern and early displays by emphasizing their shared communication purpose.
25. TACHS-R4-025 - Answer D: Position indicators made hidden elevator movement clearer and more useful to people.
The passage traces how indicators communicate elevator location to waiting riders and workers.
26. TACHS-R4-026 - Answer C: Careful research and community knowledge can restore an overlooked part of history.
Eli combines documentary research with information from a visitor, allowing the exhibit to identify a person whose story had been overlooked.
27. TACHS-R4-027 - Answer A: "First route, 1948."
The passage states that the penciled note read "First route, 1948."
28. TACHS-R4-028 - Answer D: He wanted to share verified facts without pretending the woman's identity was certain.
Eli had verified the bakery and date but not the woman's identity, so the temporary card separated known facts from uncertainty.
29. TACHS-R4-029 - Answer B: encouraged participation from.
The open question encouraged community members to contribute information.
30. TACHS-R4-030 - Answer B: Reflected sounds that return quickly and blend with the original sound
The second paragraph defines reverberation as reflections that return rapidly and merge with the original sound.
31. TACHS-R4-031 - Answer D: It explains sound reflection, distinguishes related effects, and describes design responses.
The passage moves from the cause of echoes to reverberation and then to methods designers use to control sound.
32. TACHS-R4-032 - Answer A: To show that the desired amount of reverberation depends on a room's purpose
The comparison shows that rooms need different sound behavior depending on whether speech clarity or musical richness is the goal.
33. TACHS-R4-033 - Answer C: scatter in several directions.
Irregular surfaces diffuse sound by scattering it rather than returning one strong reflection.
34. TACHS-R4-034 - Answer D: How flexible design allows public squares to support many forms of shared activity
The passage explains that squares succeed by supporting different activities at different times.
35. TACHS-R4-035 - Answer B: Trees, steps, benches, and storefronts
The second paragraph specifically names trees, steps, benches, and storefronts as useful edge features.
36. TACHS-R4-036 - Answer C: Movable seating that can be rearranged for different uses
Movable seating directly supports the author's emphasis on flexible, changing uses.
37. TACHS-R4-037 - Answer A: To emphasize that the space should support changing activities rather than function as a fixed display
A framework supports many possible activities; it is not a single fixed scene or display.
38. TACHS-R4-038 - Answer D: able to serve changing needs.
In context, adaptable means capable of serving different activities and needs.
39. TACHS-R4-039 - Answer C: One pulley wheel barely turns and the rope has shifted toward its rim.
Ms. Patel observes a nearly stuck pulley wheel and a rope displaced from the center groove.
40. TACHS-R4-040 - Answer A: To confirm that the repair works without risking the painted cloth
The sandbag allows the crew to test the repaired system while protecting the backdrop.
41. TACHS-R4-041 - Answer D: A small source of resistance can matter more than additional force.
The backdrop rises after the crew reduces friction at one pulley rather than pulling harder.
42. TACHS-R4-042 - Answer B: made stable and safe.
The ladder is made stable and safe before Ms. Patel climbs it.
43. TACHS-R4-043 - Answer B: By moving backward in a circle and flicking grains away
The first paragraph says the larva circles backward while throwing grains outward.
44. TACHS-R4-044 - Answer D: To emphasize that the trap depends on the behavior of unstable grains
The clarification highlights that sliding sand, not a mechanical pulling force, makes the trap effective.
45. TACHS-R4-045 - Answer A: It explains why the pit must have a carefully balanced slope.
The paragraph contrasts slopes that are too shallow or too steep and explains the useful balance.
46. TACHS-R4-046 - Answer C: basic ideas that explain how something works.
The word refers to general ideas about movement on unstable materials.
47. TACHS-R4-047 - Answer D: Rearrange wording so it sounds natural in the new language
The passage says a second draft may reorganize wording to sound natural.

48. TACHS-R4-048 - Answer B: To judge how naturally the wording and tone sound
Reading aloud helps a translator evaluate flow, voice, and clarity in the new language.
49. TACHS-R4-049 - Answer C: A finished translation may conceal extensive testing of alternatives.
Readers see one final wording even though the translator evaluated many alternatives.
50. TACHS-R4-050 - Answer A: Multiple drafts help translators balance meaning, clarity, and tone.
The passage explains how comparing drafts helps preserve meaning while improving natural expression and voice.

Written Expression

51. TACHS-WE4-001 - Answer B: Sentence 4

The cafeteria menu is unrelated to creating and maintaining the hallway art gallery.

52. TACHS-WE4-002 - Answer C: To help more people notice student artwork, students at Monroe Middle School proposed turning an empty hallway into a rotating gallery.

This revision clearly connects the students' purpose to their proposal in one complete sentence.

53. TACHS-WE4-003 - Answer A: concise

"Concise" precisely means brief while maintaining an appropriate formal tone.

54. TACHS-WE4-004 - Answer D: This schedule will allow many classes to share the limited wall space.

The added sentence explains why rotating the work every six weeks benefits the school.

55. TACHS-WE4-005 - Answer B: Sentence 1

Sentence 1 introduces both the question being investigated and the shade sail being tested.

56. TACHS-WE4-006 - Answer A: Although one table was covered by the sail, the other remained in direct sunlight.

"Although" correctly subordinates the first complete idea and connects it to the contrasting condition.

57. TACHS-WE4-007 - Answer D: On each clear-day trial, most readings at the shaded table were lower than those at the unshaded table.

This revision states the comparison and testing conditions precisely without vague wording.

58. TACHS-WE4-008 - Answer A: The club presented its results to the principal and recommended testing longer across different seasons.

The coordinated verbs "presented" and "recommended" are parallel and share the subject "club."

59. TACHS-WE4-009 - Answer B: Neither of the two routes leads directly to the museum.

The singular subject "neither" requires the singular verb "leads."

60. TACHS-WE4-010 - Answer C: The coach gave Elena and me the practice schedule.

"Elena and me" correctly uses the objective case for the people receiving the schedule.

61. TACHS-WE4-011 - Answer D: Because the bridge was closed, we took another route.

This sentence expresses the cause and result directly without redundancy or unnecessary passive construction.

62. TACHS-WE4-012 - Answer C: Line 3

The verb "handled" must be modified by the adverb "carefully," not the adjective "careful."

63. TACHS-WE4-013 - Answer B: Line 4

The past-time context requires "returned," rather than the present-tense form "return."

64. TACHS-WE4-014 - Answer C: One bin held cups, one held bowls, and one held divided trays.

The added sentence directly explains the three labeled container categories introduced in sentence 3.

65. TACHS-WE4-015 - Answer A: During the first lunch period, several students paused because the labels were too small to read quickly.

Choice A clearly connects the students' pause with the labels' limited readability in a complete sentence.

66. TACHS-WE4-016 - Answer D: After the larger labels were added, students sorted containers with fewer delays.

Choice D identifies the change and states its observable effect precisely.

67. TACHS-WE4-017 - Answer B: The trial showed that a return system works best when its categories are immediately clear.

Choice B connects the test results to the passage's central lesson about clear directions.

68. TACHS-WE4-018 - Answer C: After sentence 3

The sentence adds another review criterion and therefore belongs after the list in sentence 3.

69. TACHS-WE4-019 - Answer B: After hearing that one speaker rushed and another paused too often, the students marked breathing points and words for emphasis.

Choice B uses a clear introductory clause and correctly identifies the students as the people who revised the script.

70. TACHS-WE4-020 - Answer A: Sentence 7

The material of the microphone stand does not contribute to the discussion of speaking practice.

71. TACHS-WE4-021 - Answer B: Each student practiced reading the same short script into a microphone.

The singular subject "each student" correctly takes the singular past-tense verb "practiced."

72. TACHS-WE4-022 - Answer C: The advisor spoke with Maya and me after the meeting.

The preposition "with" requires the objective pronoun "me."

73. TACHS-WE4-023 - Answer A: Neither of the two signs is visible from the doorway.
The singular subject “neither” requires the singular verb “is.”
74. TACHS-WE4-024 - Answer D: Line 4
Line 4 contains the error: the singular subject “one” requires “requires.”
75. TACHS-WE4-025 - Answer C: Line 3
Line 3 contains the error: the adjective “clear,” not the adverb “clearly,” should follow “were.”
76. TACHS-WE4-026 - Answer D: Line 4
Line 4 contains the error: the plural possessive form should be “guides’ answers.”
77. TACHS-WE4-027 - Answer B: Sentence 4
The school mascot is unrelated to converting and operating the podcast booth.
78. TACHS-WE4-028 - Answer C: Our media club proposed turning an unused storage room into a compact recording booth for student projects.
Choice C identifies the club, the room, the proposed change, and the intended use clearly and precisely.
79. TACHS-WE4-029 - Answer A: After selecting removable sound-absorbing panels for the measured room, the club created a reservation calendar and operating instructions.
Choice A combines the related planning actions with clear sequencing and without a misplaced modifier.
80. TACHS-WE4-030 - Answer D: The calendar will limit each reservation to forty minutes so that more groups can use the booth.
The reservation limit directly explains how the calendar will help groups share the booth.
81. TACHS-WE4-031 - Answer C: A fair comparison requires researchers to keep most conditions the same while changing only the factor being tested.
Choice C introduces the principle of controlled comparison that the experiment illustrates.
82. TACHS-WE4-032 - Answer B: As a result,
Sentence 5 states a consequence of the unequal sunlight described in sentence 4, so “As a result” is appropriate.
83. TACHS-WE4-033 - Answer D: The club repeated the test with both containers on the same shaded table.
Choice D expresses the complete idea directly without unnecessary repetition or passive phrasing.
84. TACHS-WE4-034 - Answer A: By correcting the unequal placement, the club produced evidence that could support a dependable comparison.
Choice A links the corrected procedure to the dependability of the evidence, reinforcing the passage’s main point.
85. TACHS-WE4-035 - Answer B: Each of the volunteers has received a schedule.
The singular subject “Each” requires the singular verb “has received.”
86. TACHS-WE4-036 - Answer D: The guide showed Carlos and me where the trail began.
“Carlos and me” correctly functions as the object of the verb “showed.”
87. TACHS-WE4-037 - Answer C: Line 3
With “neither...nor,” the verb agrees with the nearer plural subject “musicians,” so the line should use “were.”
88. TACHS-WE4-038 - Answer A: Line 1
Because the model belongs jointly to two students, the plural possessive should be “students’,” not “student’s.”
89. TACHS-WE4-039 - Answer B: The photographs showed that glare and low contrast hid many numbers.
The sentence explains what the photographs revealed and leads logically to the color comparison.
90. TACHS-WE4-040 - Answer A: During a passing period, we also tested two number sizes.
The correct revision is a complete sentence with a clear subject, verb, and modifier placement.
91. TACHS-WE4-041 - Answer A: While walking through the hall, students correctly read the larger numbers more often.
Choice A clearly identifies who read the numbers, under what condition, and with what result.
92. TACHS-WE4-042 - Answer C: Sentence 7
The total number of lockers does not support the comparison of number color, size, and placement.
93. TACHS-WE4-043 - Answer A: After a cardboard grid straightened the moving air, we suspended paper shapes from a thin rod inside the box.
Choice A clearly connects the two setup steps without a run-on or misplaced modifier.
94. TACHS-WE4-044 - Answer C: At the same fan setting, the narrow triangle swung through a smaller distance than the wide rectangle.
Choice C states the testing condition and the measured difference precisely.
95. TACHS-WE4-045 - Answer B: We plan to repeat the test with shapes cut to equal areas.
The correct revision uses the standard verb form and clearly describes the planned test.

96. TACHS-WE4-046 - Answer B: The boxes that were delivered today are in the office.

The plural subject "boxes" correctly takes "were" and "are."

97. TACHS-WE4-047 - Answer C: Walking into the lab, Maya noticed that the experiment was already running.

Choice C clearly identifies Maya as the person walking into the lab.

98. TACHS-WE4-048 - Answer D: Line 4

Line 4 contains the error: the singular subject "each" requires "was."

99. TACHS-WE4-049 - Answer D: Line 4

Line 4 contains the error: the adverb "carefully" should modify "demonstrated."

100. TACHS-WE4-050 - Answer D: Line 4

Line 4 contains the error: the plural possessive form should be "editors' comments."

Mathematics

101. TACHS-M4-001 - Answer A: $6\frac{3}{4}$

$2\frac{1}{4} \times 3 = \frac{9}{4} \times 3 = \frac{27}{4} = 6\frac{3}{4}$.

102. TACHS-M4-002 - Answer B: 0

Evaluate parentheses and multiplication first: $4 + 2 = 6$, $3 \times 6 = 18$, and $18 - 18 = 0$.

103. TACHS-M4-003 - Answer C: 40%

$36 \div 90 = 0.4$, and 0.4 equals 40%.

104. TACHS-M4-004 - Answer D: 9

Divide by 5 to get $x - 2 = 7$, then add 2 to obtain $x = 9$.

105. TACHS-M4-005 - Answer A: $5n - 1$

The sequence increases by 5, and $5(1) - 1 = 4$, the first term.

106. TACHS-M4-006 - Answer C: $c = 2m + 6$

The variable charge is \$2 times m movies, and the fixed \$6 fee is then added.

107. TACHS-M4-007 - Answer D: 6

The values total 30, and $30 \div 5 = 6$.

108. TACHS-M4-008 - Answer B: $\frac{1}{16}$

The chance of blue on each draw is $\frac{3}{12} = \frac{1}{4}$, so $\frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$.

109. TACHS-M4-009 - Answer A: 62°

Supplementary angles total 180° , so $180^\circ - 118^\circ = 62^\circ$.

110. TACHS-M4-010 - Answer B: 36 cm

A square with area 81 cm^2 has side length 9 cm, so its perimeter is $4 \times 9 = 36\text{ cm}$.

111. TACHS-M4-011 - Answer C: 12 miles per hour

Average speed equals distance divided by time: $18 \div 1.5 = 12$ miles per hour.

112. TACHS-M4-012 - Answer D: 27 feet

Multiply the model length by the scale: $4.5 \times 6 = 27$ feet.

113. TACHS-M4-013 - Answer D: 13

The absolute values are 18 and 5, so $18 - 5 = 13$.

114. TACHS-M4-014 - Answer B: $\frac{11}{12}$

Using twelfths gives $\frac{10}{12} - \frac{3}{12} + \frac{4}{12} = \frac{11}{12}$.

115. TACHS-M4-015 - Answer A: 9

Since $36 = 2^2 \times 3^2$, it has $(2 + 1)(2 + 1) = 9$ positive factors.

116. TACHS-M4-016 - Answer C: 3

The rate of change is $(17 - 5) \div (6 - 2) = 12 \div 4 = 3$.

117. TACHS-M4-017 - Answer D: $10x - 15$

Distribute and combine like terms: $6x - 15 + 4x = 10x - 15$.

118. TACHS-M4-018 - Answer A: 3.6

The total is $2 + 9 + 20 + 5 = 36$ across 10 values, so the mean is 3.6.

119. TACHS-M4-019 - Answer B: $\frac{1}{12}$

Only A is a vowel and only 2 is even, so the probability is $\frac{1}{4} \times \frac{1}{3} = \frac{1}{12}$.

120. TACHS-M4-020 - Answer D: The data show a positive association between practice time and points.

The point values generally rise as practice hours increase, showing a positive association.

121. TACHS-M4-021 - Answer C: (6, 7)

The missing vertex must share $x = 6$ with one point and $y = 7$ with another, giving (6, 7).

122. TACHS-M4-022 - Answer B: $\frac{1}{4}$

The sector represents $\frac{90}{360}$ of the circle, which simplifies to $\frac{1}{4}$.

123. TACHS-M4-023 - Answer A: 141.3 cubic centimeters

Volume is $\pi r^2 h = 3.14 \times 3^2 \times 5 = 141.3$ cubic centimeters.

124. TACHS-M4-024 - Answer C: 6,000 milliliters

The rate is 1.5 liters per minute, so 4 minutes gives 6 liters, or 6,000 milliliters.

125. TACHS-M4-025 - Answer D: 25,000
Multiply 2.5 by 10,000 to get 25,000 square centimeters.
126. TACHS-M4-026 - Answer D: $1\frac{3}{4}$ cups
The unit rate is $\frac{3}{4} \div 6 = \frac{1}{8}$ cup per serving; $14 \times \frac{1}{8} = \frac{14}{8} = 1\frac{3}{4}$ cups.
127. TACHS-M4-027 - Answer A: \$92
Fifteen percent of \$80 is \$12, so the new price is $\$80 + \$12 = \$92$.
128. TACHS-M4-028 - Answer C: 0
 $-4 + 13 = 9$, and $9 - 9 = 0$.
129. TACHS-M4-029 - Answer B: 6
Subtract 7 to get $4x = 24$, then divide by 4 to get $x = 6$.
130. TACHS-M4-030 - Answer A: 162
Each term is multiplied by 3, so $54 \times 3 = 162$.
131. TACHS-M4-031 - Answer D: 1
The mean is $(6 + 8 + 8 + 10 + 13) \div 5 = 9$, and the median is 8; the difference is 1.
132. TACHS-M4-032 - Answer B: $\frac{2}{3}$
There are 12 tiles total and 8 are not blue, so the probability is $\frac{8}{12} = \frac{2}{3}$.
133. TACHS-M4-033 - Answer C: 77°
An exterior angle equals the sum of the two remote interior angles, so $124^\circ - 47^\circ = 77^\circ$.
134. TACHS-M4-034 - Answer D: 68 square feet
The large rectangle has area 80 square feet, and the removed rectangle has area 12 square feet; $80 - 12 = 68$.
135. TACHS-M4-035 - Answer A: 140 cm^3
Volume = length $\times$ width $\times$ height = $7 \times 5 \times 4 = 140$ cubic centimeters.
136. TACHS-M4-036 - Answer C: 275
One meter equals 100 centimeters, so $2.75 \times 100 = 275$ centimeters.
137. TACHS-M4-037 - Answer B: 6 units
The points have the same y-coordinate, so the horizontal distance is $4 - (-2) = 6$ units.
138. TACHS-M4-038 - Answer C: $\frac{1}{2}$
The average of $\frac{2}{5}$ and $\frac{3}{5}$ is $\frac{5}{10}$, which equals $\frac{1}{2}$.
139. TACHS-M4-039 - Answer D: 72
Compute $2^3 \times 3^2 = 8 \times 9 = 72$.
140. TACHS-M4-040 - Answer A: 7
The next multiple of 9 after 587 is 594, and $594 - 587 = 7$.
141. TACHS-M4-041 - Answer B: (3, -1)
Substitute $y = 2 - x$: $2 - x = -2x + 5$, so $x = 3$ and $y = -1$.
142. TACHS-M4-042 - Answer C: -8
Increasing x by 3 changes y by -9, so $y = 1 - 9 = -8$.
143. TACHS-M4-043 - Answer A: The mean changes more.
The outlier raises the mean substantially, while the median changes only from 6 to 6.5.
144. TACHS-M4-044 - Answer D: $\frac{9}{20}$
There are 6 multiples of 3 and 4 multiples of 5, with 15 counted twice, so 9 of 20 numbers qualify.
145. TACHS-M4-045 - Answer C: 14.4 minutes
The total time is $3 \times 12 + 2 \times 18 = 72$ minutes, and $72 \div 5 = 14.4$ minutes.
146. TACHS-M4-046 - Answer B: 7 centimeters
Since area = base $\times$ height, the height is $84 \div 12 = 7$ centimeters.
147. TACHS-M4-047 - Answer D: 35 square units
The side lengths are 7 and 5 units, so the area is $7 \times 5 = 35$ square units.
148. TACHS-M4-048 - Answer A: (-4, -3)
Reflection across the y-axis changes the sign of x while leaving y unchanged.
149. TACHS-M4-049 - Answer B: \$2.50
Divide the total cost by the weight: $\$8.75 \div 3.5 = \2.50 per pound.

150. TACHS-M4-050 - Answer C: 12 liters
Multiply 0.012 by 1,000 to get 12 liters.

Ability

151. TACHS-A4-001 - Answer D: ■■■

Moving right adds two copies, and moving down changes the open circle family to filled squares.

152. TACHS-A4-002 - Answer A: ↑

Rotating ← 90° clockwise produces ↑, continuing the pattern across both the row and column.

153. TACHS-A4-003 - Answer B: ◇ inside △

The inner and outer shapes exchange positions across each row without changing their forms.

154. TACHS-A4-004 - Answer C: △▲

Each third cell combines the first and second symbols in the same left-to-right order.

155. TACHS-A4-005 - Answer A: filled ▲ above filled ◆

Both bottom-row shapes keep their positions and change from open to their corresponding filled forms.

156. TACHS-A4-006 - Answer A: Near both lower corners, symmetric across the vertical centerline

Undoing the horizontal half-fold reflects the lower-left punch to the lower-right side.

157. TACHS-A4-007 - Answer B: Four holes, one in each quadrant, symmetric across both centerlines

Reversing two perpendicular folds reflects the punch across both centerlines, producing four symmetric locations.

158. TACHS-A4-008 - Answer C: A symmetric diamond-shaped opening centered on the diagonal

The triangular notch on the fold reflects across the diagonal to form a symmetric diamond-shaped opening.

159. TACHS-A4-009 - Answer D: 4

Reversing two perpendicular folds reflects the interior punch across both centerlines, producing four holes.

160. TACHS-A4-010 - Answer B: ☆|☆

Each example places two identical open symbols on opposite sides of a vertical line.

161. TACHS-A4-011 - Answer C: ◇◆

Each pair contains an open shape followed by the filled form of that same shape.

162. TACHS-A4-012 - Answer D: ◇ above ○

Like each example, the choice places two different open symbols in a vertical arrangement.

163. TACHS-A4-013 - Answer A: ↑ ■

Rotating ← clockwise gives ↑, and changing □ to its filled form gives ■.

164. TACHS-A4-014 - Answer C: ■■

Five squares minus three squares leaves two, and the result is filled: ■■.

165. TACHS-A4-015 - Answer D: □□

The third row continues the shift with △△△, ○, then □□.

166. TACHS-A4-016 - Answer B: Bottom-right ■

The bottom-left open square shifts to bottom-right and changes to a filled square.

167. TACHS-A4-017 - Answer D: Four holes in a 2 × 2 arrangement

Two perpendicular half-folds create four layers, so the punch reflects into four quadrant positions.

168. TACHS-A4-018 - Answer A: Four holes in one horizontal row

Each of the four stacked panels receives the punch, producing one hole in each panel across the sheet.

169. TACHS-A4-019 - Answer B: Two holes reflected across the diagonal

A single diagonal fold creates one reflected partner across that diagonal.

170. TACHS-A4-020 - Answer C: One circular hole centered on the vertical midline

The semicircle lies against the fold, so its reflected half forms one complete circular hole on the midline.

171. TACHS-A4-021 - Answer A: ◆◇◆

Each group places one open shape between two filled copies of that same shape.

172. TACHS-A4-022 - Answer D: ◇ \ ◇

Each valid group places one line between two identical open shapes.

173. TACHS-A4-023 - Answer C: ◆◇◇

Each arrangement contains one filled shape followed by two open copies of that same shape.

174. TACHS-A4-024 - Answer B: ← ↑ →

Each group lists three consecutive directions in clockwise order.

175. TACHS-A4-025 - Answer A: ■

Across the row, the open shape becomes its filled counterpart; therefore, the open square becomes a filled square.

176. TACHS-A4-026 - Answer C: ○

Each row and column must contain one of each shape, so the missing symbol is a circle.

177. TACHS-A4-027 - Answer B: ← □□

From top to bottom, the right arrow becomes a left arrow and circles become squares; the two-shape count remains unchanged.

178. TACHS-A4-028 - Answer D: ▲ inside ◇

Only the inner shape changes to its filled form, so the open triangle becomes a filled triangle inside the same diamond.

179. TACHS-A4-029 - Answer C: ↑

Continuing the 90° counterclockwise rotation after the right arrow produces an up arrow.

180. TACHS-A4-030 - Answer A: Two holes near the upper-left and upper-right corners, mirrored across the vertical centerline

Undoing one vertical fold reflects the punch across the vertical centerline, producing two matching upper holes.

181. TACHS-A4-031 - Answer D: Four holes, one in each quadrant, symmetric across both centerlines

Two perpendicular folds create four layers, so the punch is reflected across both centerlines.

182. TACHS-A4-032 - Answer B: One symmetric diamond-shaped opening on the diagonal

Because the cut lies on the diagonal fold, the two triangular halves join into one symmetric diamond-shaped opening.

183. TACHS-A4-033 - Answer C: Two matching semicircular notches on the left edge, mirrored across the horizontal centerline

Undoing the horizontal fold reflects the edge notch to a second position on the same left edge.

184. TACHS-A4-034 - Answer D: ☆ ★ ☆

Each valid arrangement has matching open outer shapes with the filled version of that shape in the center.

185. TACHS-A4-035 - Answer A: ← ↑ →

Each arrow is obtained by rotating the previous arrow 90° clockwise.

186. TACHS-A4-036 - Answer B: ◇ inside △; ◆ outside △

The rule places an open shape inside a different open boundary and its filled counterpart outside that same boundary.

187. TACHS-A4-037 - Answer C: ■ ● ■ ●

Each valid arrangement alternates two symbols in an A-B-A-B pattern.

188. TACHS-A4-038 - Answer D: ■ →

Filling □ gives ■, and rotating ↑ clockwise gives →.

189. TACHS-A4-039 - Answer A: ▲

Four triangles minus three triangles leaves one, which becomes filled: ▲.

190. TACHS-A4-040 - Answer C: □

The repeating left shift makes the final symbol a square.

191. TACHS-A4-041 - Answer B: Top □; Bottom ■

A clockwise rotation places the left open square on top and the right filled square on the bottom.

192. TACHS-A4-042 - Answer D: Four holes in a 2 × 2 arrangement

Two perpendicular half-folds create four reflected hole positions.

193. TACHS-A4-043 - Answer B: Two matching notches reflected across the diagonal

The outer-edge cut is duplicated by reflection across the diagonal fold.

194. TACHS-A4-044 - Answer A: Three holes in a vertical column near the left side

All three horizontal strips are stacked, so the punch appears once in each strip at the same left position.

195. TACHS-A4-045 - Answer D: One circular hole centered on the horizontal midline

The two semicircular halves reflect across the fold to form one complete circle on the midline.

196. TACHS-A4-046 - Answer B: ◇ ◆ ◇

Each valid group has matching open shapes on the outside and the filled version in the center.

197. TACHS-A4-047 - Answer A: ← ◇◇

Each arrangement has one direction arrow followed by two identical open shapes.

198. TACHS-A4-048 - Answer C: ◇ \ ◆

Each group contains an open shape, a line, and the matching filled shape in that order.

199. TACHS-A4-049 - Answer C: $\leftarrow \uparrow \downarrow$

In each group, the second arrow is 90° clockwise from the first and the third is 90° counterclockwise.

200. TACHS-A4-050 - Answer D: Open triangle $\triangle$ containing $\blacklozenge$

Each figure has an open boundary shape containing a different filled shape.